Assure: Home Care Assistant

Wavve: Contactless Sleep Sensor

Wavve Hub & Kit: Elderly Home Care System

Our-cloud-restful-api

V2.7

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HISTORY

Date	Version	Modification
04-06-2023	V1.0	Release.
04-12-2023	V1.1	Change interface description of section 2.3.
05-29-2023	V1.2	Update room information, zigbee network, ARM/DISARM/DELAY SETTING etc.
8-08-2023	V1.3	Edit enumeration type, the time-zone of the Room. New interface to obtain reports based on Radar ID.
10-11-2023	V1.4	Correct sleep report -1.
10-23-2023	V1.5	New report type: the Sleep score. Only one Wavve for the same user.
11-1-2023	V1.6	Emphasis on the time zone setting of the room (3.20). Add firmware update and status (3.35-3.36) Add machine learning (3.37-3.38)
01-11-2024	V1.7	To learn the room interference (3.39) In-bed detection for the Assure (2.4)
03-08-2024	V1.8	To define the alarm type of the new add-on
03-21-2024	V1.9	Added door and bed position learning in the room
08-02-2024	V2.0	To request for the room layout assigned to a specific radar (3.41)~ To get the installation mode (fall-detection mode) of the radar (3.44)
11-26-2024	V2.1	Add radar type; delete bed-exit and no bio-vital alert of the Wavve; new time format that complies with ISO-8601.
02-24-2025	V2.2	updateDefinition of Alert Type (Function id); Add the add-on type;Add Error Code 406001051. etc
07-22-2025	V2.3	Add alert Type (Function id: 101 End of report)
08-25-2025	V2.4	Add Error code; Add interface to learn the environmental interference.
09-10-2025	V2.5	Change interface description of section 3.22 and 3.33.
11-25-2025	V2.6	Add 3 new error codes; Replace the installation height interface with separate interfaces to set the installation height for the Assure

		and the Wavve.
12-01-2025	V2.7	Real-time target coordinates: To get the push switch settings of the system (real-time coordinates of the target) (3.48). To set the push switch settings of the system (real-time coordinates of the target) (3.49).

Contents

1. Introduction

Codes in the format below will be returned when the API is requested. When code is 200, it means the request is successful, otherwise unsuccessful. In the case of a failed request, the error message will be shown in msg. Please refer to 2.5 for more information.

```
{
        "code": "200",
        "data": {
                ...
        },
        "msg": "success"
}
```

2. Common Enumeration Type

2.1. Definition of Radar Model (radar type)

Radar type	Value
Assure	0
Wavve | Wavve hub	1

2.2. Definition of Room Type

Room type	Value
Living Room	1
Bedroom	2
Bathroom	3
Kitchen	4
Office	5
Others	6

2.3. Definition of Alert Type (Function id)

Function id	Value	Description	Type	Radar typeDevices	RoomTypes
Oversleeping Alert	1	To learn the user's sleep habit and get alerted if the user is still in bed beyond his normal schedule.	Alert	Wavve, Wavve hub, Assure	2

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Function id	Value	Description	Type	Radar typeDevices	RoomTypes
Frequent Night Activity	2	Number of night activities at night. For the Assure, it covers the time between 23:00 pm to 6:00am the next day. For the Wavve, it covers the time between the user goes to sleep at wakes up the next morning.	Remind	Wavve, Wavve hub, Assure	2
Wandering Alert (Night)	3	To get an alert if the user is absent from the room (Assure) or the bed (Wavve) longer than expected at night. For the Assure, it covers the time between 18:00 pm to 6:00am the next day. For the Wavve, it covers the time between the user goes to sleep and wakes up the next morning.	Remind&Alert	Wavve,Wavve hub, Assure	2
Abnormal Sleep	4	To get alerts if the user is still up past the set bedtime.	Remind	Wavve, Wavve hub, Assure	2
Fall alert	5	Fall management with learning algorithm.	Alert	Assure	all
Intrusion alert	6	The radar will send an intrusion alert whenever a moving target is detected when the surveillance mode is activated.	Alert	Assure	all
Wandering	7	To get an alert if the	Remind&Alert	Assure	1,2,3

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Function id	Value	Description	Type	Radar typeDevices	RoomTypes
(Day)		user is absent from the room longer than expected during the day (24h).			
Continuous Sedentary Notification	8	To analyze the user's continuous sedentary behavior and accumulative sedentary behavior within a 24-hour period and get alerts if the user is sitting longer than expected.	Remind&Alert	Assure	1,2,4,5
Accumulative Sedentary Notification	9	To analyze the user's continuous sedentary behavior and accumulative sedentary behavior within a 24-hour period and get alerts if the user is sitting longer than expected.	Remind	Assure	1,2,4,5
Nutrition Alert	11	To see if the user has been using the kitchen regularly as he should be.	Remind	Assure	4
Extended Toileting	12	Get alerts if the user has been in the toilet for too long.	Remind&Alert	Assure	3
Frequent Toileting (Night)	13	Get alerts if the user has been using the toilet too often at night (23:00pm - 6:00am).	Remind	Assure	3
SOS alarm	21	SOS	Alert	Add-ons (Wavve hub)	all
Tamper alarm	22	Tamper alert, applicable to devices with tamper alarm	Alert	Add-ons (Wavve hub)	all

Function id	Value	Description	Type	Radar typeDevices	RoomTypes
		only.			
Low battery notification	23	Low battery alert	Alert	Add-ons (Wavve hub)	all
Smoke alarm	26	Smoke alert	Alert	Add-ons (Wavve hub)	all
Water leakage	27	Water leakage alert	Alert	Add-ons (Wavve hub)	all
CO alarm	28	CO alert	Alert	Add-ons (Wavve hub)	all
Gas leakage	29	Gas alert	Alert	Add-ons (Wavve hub)	all
PIR motion alert	30	Whenever a valid target is detected, the alert will be triggered. The alert will be cleared only when the filed of view is empty for longer than 1 min. The alert and cancellation are pushed via the **/api/alarm** and **/api/eliminate-alarm** interfaces, respectively.	Alert&Eliminate	Add-ons (Wavve hub)	all
PIR motion alert- and tamper alert	31	Get alert if the device is tempered with.	Alert	Add-ons (Wavve hub)	all
End of report	101	A notification will be pushed via the /api/alarm endpoint to mark the end of the report. (Note: only applicable to the sleep report.)	Remind	Wavve,Wavve hub	2

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2.4. Definition of Report Type

Report type	Value	Description	Radar type	Room type
sleep delete	-1	The total time of device offline and night activities during sleep.	1	2
offline	0	The time that the device is offline.	1	2
Awake	1	When the user is awake.	1	2
Rem	2	Sleep stages: REM	1	2
light sleep	3	Sleep stages: light sleep	1	2
deep sleep	4	Sleep stages: deep sleep	1	2
out of bed	5	When the user is out of bed (night activity) during the Sleep.	1	2
breath value	6	The average breath rate within a certain period.	1	2
heart value	7	The average heart rate within a certain period.	1	2
sleep	8	The total time between the user falls asleep and wakes up the next day. (The actual user sleep time is Value-1 minus Value8.)	1	2
sleep score	9	The sleep score.	1	2
sedentary	10	The time when the user is sitting.	0	1,2,4,5
Activity	11	The time when the user is neither sleeping, nor sitting down.	0	1,2,4,5
out of	12	The time when there is no moving target	0	1,2,3,4,5

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Report type	Value	Description	Radar type	Room type
room		detected within the room.		
bathroom use	13	The time when the bathroom is occupied.	0	3
In-bed	14	To detect the time when the user is in bed.	0	2

2.5. Error Codes

Error code	Description
200	success
406	Fail
401	unauthenticated
403	forbidden
404	Not Found
405	Method Not Allowed
500	Internal Server Error
406000001	Parameter verification error
406000002	Parameter type error
406000003	Data error
406000004	Server busy
406001001	Radar offline
406001003	Radar is not assigned to a room
406001004	Unable to pair the radar
406001005	This device has been assigned to this room
406001006	Wavve must be assigned to a bedroom
406001007	A radar with the same model has been assigned to this room.
406001008	The room does not exist.
406001009	The radar does not exist.

Error code	Description
406001010	Failed to communicate to the radar.
406001011	App not exist.
406001012	Failed to send the results of the algorithm.
406001013	The learning of the Door is still in progress.
406001014	The Function type doesn’t match to the Room type.
406001015	The radar height is invalid. The radar must be installed within the range shown in the user manual.
406001017	Please set the time zone for the room first
406001019	App is disabled
406001020	To delete the device, please turn it off and make sure it’s offline.
406001030	Add-on detected
406001031	Failed to communicate with the add-on
406001032	The log to setup the ARM of the room does not exist
406001033	The log to setup the DELAY SETTING of the room does not exist
406001034	Failed to turn on the Zigbee network of the radar
406001035	Failed to turn off the Zigbee network of the radar
406001036	The log to setup the DELAY SETTING of the room already exists.
406001037	The TIME must be after the current time
406001038	Invalid radar type
406001039	The installation mode is not supported.
406001041	There can be only one Wavve registered with one user.
406001045	No radar firmware available for upgrade.
406001046	The door must contain 2 coordinate points.
406001047	A room only supports up to 4 doors.
406001048	The bed must contain 4 coordinate points.
406001049	Your radar firmware version does not support this feature
406001050	The time zone of the time parameter must be the same as the time zone of the room.

Error code	Description
406001051	Failed to turn off the Zigbee network of the add-on.
406001053	Radar does not support this function
406001054	The alert setting must be greater than the set value.setting.
406001055	The working distance of theWavve radar must be between 0.9m and 3m.
406001056	The working distance of the Assure radar must be between 1m and 7m.

2.6. Type of the add-ons

Add-on type	Description
1	Door Sensor
2	SOS Button
3	Indoor Siren
4	Smoke Detector
5	Water Detector
6	CO Detector
7	GAS Detector
8	Pull rope call button
9	Wireless PIR Motion Detector
10	Wireless PIR Motion Detector(adjustable)

3. Cloud-to-Cloud RESTful APIs

3.1. To get API information

Url:/api/v1/api-info

Method: GET

Produces: application/x-www-form-urlencoded

Consumes: */*

Note: to get the API information of the current system

Request parameter:

Parameter	Description	In	Required?	Data type	schema
Authorization	Api access token	header	true	string	

Response status:

Status code	Description	schema
200	OK	restful api response object«Api info DTO»
401	Unauthorized	
403	Forbidden	
404	Not Found	

Response parameter:

Parameter	Description	Type	schema
code	response code	string	
data	response data	Api info DTO	Api info DTO
apiVersion	api version	string	
msg	response description message	string	

Response Example:


```
{
        "code": "200",
        "data": {
                "apiVersion": ""
        },
        "msg": "success"
}
```


3.2. To get the server information

Url:/api/v1/server-info

Method: GET

Produces: application/x-www-form-urlencoded

Consumes: */*

Note: to get the current server information, including the time and time zone.

Request parameter:

Parameter	Description	In	Required?	Data type	schema

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Parameter	Description	In	Required?	Data type	schema
Authorization	Api access token	header	true	string	

Response Status:

Status code	Description	schema
200	OK	restful api response object«Server info dto»
401	Unauthorized	
403	Forbidden	
404	Not Found	

Response Parameter:

Parameter	Description	Type	schema
code	response code	string	
data	response data	Server info dto	Server info dto
serverTime	Server current time(ISO-8601)	string(date-time)	
timeZone	Server time zone	string	
msg	response description message	string	

Response Example:

```
{
        "code": "200",
        "data": {
                "serverTime": "",
                "timeZone": ""
        },
        "msg": "success"
}
```

Response Example:

```
{
        "code": "200",
        "msg": "success"
}
```

3.3. To get the radar information

Url:/api/v1/radar-info/{radarId}

Method: GET

Produces: application/x-www-form-urlencoded

Consumes: */*

Note: Once the radar is successfully connected to AxEnd's cloud, the developer can obtain the radar information via radarID.

Request parameter:

Parameter	Description	In	Required?	Data type	schema
Authorization	Api access token	header	true	string	
radarId	radarId	path	true	string	

Response Status:

Status code	Description	schema
200	OK	restful api response object«Radar information dto»
401	Unauthorized	
403	Forbidden	
404	Not Found	

Response Parameter:

Parameter	Description	Type	schema
code	response code	string	
data	response data	Radar information dto	Radar information dto
firmwareVersion	radar firmware version	string	
online	radar is online	boolean	
radarAddress	radar connect ip address	string	
radarId	radar id	string	
radarType	radar type	integer(int32)	
serverTime	current server date time	string(date-time)	
zonedServerTime	Time in the format of ISO-8601	string(date-time)	
assignedRoomId	ID of the room that the radar is assigned to.	integer(int64)	
msg	response description message	string	

Response Example:

```
{
        "code": "200",
        "data": {
                "firmwareVersion": "",
                "online": false,
                "radarAddress": "",
                "radarId": "",
                "radarType": 0,
                "serverTime": "",
                "zonedServerTime": "",
                "assignedRoomId": 0
                },
        "msg": "success"
}
```


3.4. To get the list of all the connected radarIDs

Url:/api/v1/radar-id-list

Method:GET

Produces:application/x-www-form-urlencoded

Consumes:*/*

Note: Once the radar is successfully connected to AxEnd's cloud, the developer can obtain a list of all the radarIDs that have been successfully connected via this interface.

Request parameter:

Parameter	Description	In	Required?	Data type	schema
Authorization	Api access token	header	true	string	

Response Status:

Status code	Description	schema
200	OK	restful api response object«Void»
201	Created	
401	Unauthorized	
403	Forbidden	
404	Not Found	

Response Parameter:

Parameter	Description	Type	schema
code	response code	string	
data	response data	AccessRadarListDTO	AccessRadarListDTO

Parameter	Description	Type	schema
radarIds	Radar id list	array	string
msg	response description message	string	

Response Example:

```
{
	"code": "200",
	"data": {
		"radarIds": []
	},
	"msg": "success"
}
```

3.5. To set the buffer time (alert time) to trigger the fall alert (Assure)

Url:/api/v1/radar/set-fall-alert-buffer-time

Method: POST

Produces: application/x-www-form-urlencoded, application/json

Consumes: */*

Note: To set the buffer time of the Assure. The buffer time is the time interval between the radar detecting a fall and sending an alert. It must be between 30-120s.

Example:

```
{
 "bufferTime": 0,
 "radarId": ""
}
```

Request parameter:

Parameter	Description	In	Required?	Data type	schema
Authorization	Api access token	header	true	string	
fall alert buffer time dto	Fall alert buffer time dto	body	true	Fall alert buffer time dto	Fall alert buffer time dto
bufferTime	Unit:s. The time interval between the radar detects a fall and sends an alert, and must be between 30s to 120s.		true	integer(int32)	
radarId	radarId		true	string	

Response Status:

Status code	Description	schema
200	OK	restful api response object«Invoke radar result»
201	Created	
401	Unauthorized	
403	Forbidden	
404	Not Found	

Response Parameter:

Parameter	Description	Type	schema
code	response code	string	
data	response data	Invoke radar result	Invoke radar result
result	Operation result	boolean	
msg	response description message	string	

Response Example:

```
{
        "code": "200",
        "data": {
                "result": false
        },
        "msg": "success"
}
```

3.6. To get the buffer time (alert time) to trigger the fall alert (Assure)

Url:/api/v1/radar/fall-alert-buffer-time/{radarId}

Method: GET

Produces: application/x-www-form-urlencoded

Consumes:*/*

Note: To get the buffer time of the Assure. The buffer time is the time interval between the radar detecting a fall and sending an alert. It must be between 30-120s.

Request parameter:

Parameter	Description	In	Required?	Data type	schema
Authorization	Api access token	header	true	string	

Parameter	Description	In	Required?	Data type	schema
radarId	radarId	path	true	string	

Response Status:

Status code	Description	schema
200	OK	restful api response object«Fall alert buffer time dto»
401	Unauthorized	
403	Forbidden	
404	Not Found	

Response Parameter:

Parameter	Description	Type	schema
code	response code	string	
data	response data	Fall alert buffer time dto	Fall alert buffer time dto
bufferTime	Unit:s. The time interval between the radar detects a fall and sends an alert, and must be between 30s to 120s.	integer(int32)	
radarId	radarId	string	
msg	response description message	string	

Response Example:


```
{
        "code": "200",
        "data": {
                "bufferTime": 0,
                "radarId": ""
        },
        "msg": "success"
}
```


3.7. To set the installation height of the Assure

Url:/api/v1/radar/install-height

Method: PUT

Produces: application/x-www-form-urlencoded, application/json

Consumes: */*

Note: To set the installation height of the Assure, thedistance from the radar to the ground.

Example:


```
{
  "installHeight": 0,
  "radarId": ""
}
```


Request parameter:

Parameter	Description	In	Required?	Data type	schema
Authorization	Api access token	header	true	string	
radarInstallParamDTO	RadarInstallParamDTO	body	true	RadarInstallParamDTO	RadarInstallParamDTO
installHeight	Unit: m. Installation height, 1.4-2.2m for the Assure		true	number(bigdecimal)	
radarId	radarId		true	string	

Response Status:

Status code	Description	schema
200	OK	restful api response object«Invoke radar result»
201	Created	
401	Unauthorized	
403	Forbidden	
404	Not Found	

Response Parameter:

Parameter	Description	Type	schema
code	response code	string	
data	response data	Invoke radar result	Invoke radar result
result	Operation result	boolean	
msg	response description message	string	

Response Example:

```
{
	"code": "200",
	"data": {
		"result": false
	},
	"msg": "success"
}
```


3.8. To get the installation height of the Assure

Url:/api/v1/radar/install-height/{radarId}

Method: GET

Produces: application/x-www-form-urlencoded

Consumes: */*

Note: To get the installation height of the Assure, the distance from the radar to the ground.

Request parameter:

Parameter	Description	In	Required?	Data type	schema
Authorization	Api access token	header	true	string	
radarId	radarId	query	true	string	

Response Status:

Status code	Description	schema
200	OK	restful api response object«Install height dto»
401	Unauthorized	
403	Forbidden	
404	Not Found	

Response Parameter:

Parameter	Description	Type	schema
code	response code	string	
data	response data	Install height dto	Install height dto
installHeight	Unit: m. Installation height, 1.4-2.2m for the Assure	number(bigdecimal)	
radarId	radarId	string	

Parameter	Description	Type	schema
msg	response description message	string	

Response Example:


```
{
        "code": "200",
        "data": {
                "installHeight": 0,
                "radarId": ""
        },
        "msg": "success"
}
```


3.9. To set the working distance of the radar (Assure/Wavve)

Url:/api/v1/radar/set-working-distance

Method: POST

Produces: application/x-www-form-urlencoded, application/json

Consumes: */*

Note: Working distance defines the largest working range. The radar will no longer report signal beyond this range.

- The working distance for Wavve must be between 0.9-3m.
- The working distance for Assure must be between 1-7m.

Example:


```
{
 "radarId": "",
 "workingDistance": 0
}
```


Request parameter:

Parameter	Description	In	Required?	Data type	schema
Authorization	Api access token	header	true	string	
working distance dto	Working distance dto	body	true	Working distance dto	Working distance dto
radarId	radarId		true	string	
workingDistance	Unit: m. Must be between 1m to 7m for the Assure, and		true	number(bigdecimal)	

Parameter	Description	In	Required?	Data type	schema
	0.9m to 3m for the Wavve				

Response Status:

Status code	Description	schema
200	OK	restful api response object«Invoke radar result»
201	Created	
401	Unauthorized	
403	Forbidden	
404	Not Found	

Response Parameter:

Parameter	Description	Type	schema
code	response code	string	
data	response data	Invoke radar result	Invoke radar result
result	Operation result	boolean	
msg	response description message	string	

Response Example:


```
{
        "code": "200",
        "data": {
                "result": false
        },
        "msg": "success"
}
```


3.10. To get the working distance of the radar (Assure/Wavve)

Url:/api/v1/radar/working-distance/{radarId}

Method: GET

Produces: application/x-www-form-urlencoded

Consumes: */*

Note: to get the working distance of the radar.

Request parameter:

Parameter	Description	In	Required?	Data type	schema
Authorization	Api access token	header	true	string	
radarId	radarId	path	true	string	

Response Status:

Status code	Description	schema
200	OK	restful api response object«Working distance dto»
401	Unauthorized	
403	Forbidden	
404	Not Found	

Response Parameter:

Parameter	Description	Type	schema
code	response code	string	
data	response data	Working distance dto	Working distance dto
radarId	radarId	string	
workingDistance	Unit: m	number(bigdecimal)	
msg	response description message	string	

Response Example:


```
{
        "code": "200",
        "data": {
                "radarId": "",
                "workingDistance": 0
        },
        "msg": "success"
}
```


3.11. To get the Room information that the radar is assigned to

Url:/api/v1/radar/assigned-room-info/{radarId}

Method: GET

Produces: application/x-www-form-urlencoded

Consumes: */*

Note: To get the information of the Room that the radar is assigned to. The response will be Error is the radar is not assigned to any Room.

Request parameter:

Parameter	Description	In	Required?	Data type	schema
Authorization	Api access token	header	true	string	
radarId	radarId	path	true	string	

Response Status:

Status code	Description	schema
200	OK	restful api response object«Room info dto»
401	Unauthorized	
403	Forbidden	
404	Not Found	

Response Parameter:

Parameter	Description	Type	schema
code	response code	string	
data	response data	Room info dto	Room info dto
roomId	room id	integer(int64)	
roomName	room name	string	
roomType	room type	integer(int32)	
timeZone	Room time zone(UTC/GMT-5)	string	
msg	response description message	string	

Response Example:


```
{
        "code": "200",
        "data": {
                "roomId": 0,
                "roomName": "",
                "roomType": 0,
                "timeZone": ""
        },
        "msg": "success"
}
```


3.12. To create a Room

Url:/api/v1/room/create

Method: POST

Produces: application/x-www-form-urlencoded,application/json

Consumes: */*

Note: To create a room, with defined type, name. The response is the Room information.

Example:

```
{
 "roomName": "",
 "roomType": 0,
"timeZone":""
}
```

Request parameter:

Parameter	Description	In	Required?	Data type	schema
Authorization	Api access token	header	true	string	
create room DTO	Create room DTO	body	true	Create room DTO	Create room DTO
roomName	room name		true	string	
roomType	room type		true	integer(int32)	
timeZone	Room time zone(UTC/GMT-5)		true	string	

Response Status:

Status code	Description	schema
200	OK	restful api response object«Room info dto»
201	Created	
401	Unauthorized	
403	Forbidden	
404	Not Found	

Response Parameter:

Parameter	Description	Type	schema
code	response code	string	
data	response data	Room info dto	Room info dto
roomId	room id	integer(int64)	

Parameter	Description	Type	schema
roomName	room name	string	
roomType	room type	integer(int32)	
timeZone	Room time zone(UTC/GMT-5)	string	
msg	response description message	string	

Response Example:


```
{
        "code": "200",
        "data": {
                "roomId": 0,
                "roomName": "",
                "roomType": 0,
                "timeZone": ""
        },
        "msg": "success"
}
```


3.13. To create a Room and assign the radar to the Room

Url:/api/v1/room/create-and-assign

Method: POST

Produces: application/x-www-form-urlencoded, application/json

Consumes: */*

Note: To create a new Room and assign a radar to this Room. Please note that the Wavve can be assigned only to the bedroom, yet the Assure can be assigned to all the different types.

Example:


```
{
  "radarId": "",
  "roomName": "",
  "roomType": 0,
"timeZone": ""
}
```


Request parameter:

Parameter	Description	In	Required?	Data type	schema
Authorization	Api access token	header	true	string	
create and assign room DTO	Create and assign room DTO	body	true	Create and assign room DTO	Create and assign room DTO

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Parameter	Description	In	Required?	Data type	schema
radarId	radar id		true	string	
roomName	room name		true	string	
roomType	room type		true	integer(int32)	
timeZone	Room time zone(UTC/GMT-5)		true	string	

Response Status:

Status code	Description	schema
200	OK	restful api response object«Room info dto»
201	Created	
401	Unauthorized	
403	Forbidden	
404	Not Found	

Response Parameter:

Parameter	Description	Type	schema
code	response code	string	
data	response data	Room info dto	Room info dto
roomId	room id	integer(int64)	
roomName	room name	string	
roomType	room type	integer(int32)	
timeZone	Room time zone(UTC/GMT-5)	string	
msg	response description message	string	

Response Example:


```
{
        "code": "200",
        "data": {
                "roomId": 0,
                "roomName": "",
                "roomType": 0,
                "timeZone": ""
        },
        "msg": "success"
}
```

3.14. To assign the radar to a Room

Url:/api/v1/room/assign

Method: POST

Produces: application/x-www-form-urlencoded, application/json

Consumes: */*

Note: To assign the radar to a Room. There can be only one radar with the same model in the same Room. Please note that the Wavve can be assigned only to the bedroom, yet the Assure can be assigned to all the different types.

Example:

```
{
 "radarId": "",
 "roomId": 0
}
```

Request parameter:

Parameter	Description	In	Required?	Data type	schema
Authorization	Api access token	header	true	string	
assign room DTO	Assign room DTO	body	true	Assign room DTO	Assign room DTO
radarId	radar id		true	string	
roomId	roomId		true	integer(int64)	

Response Status:

Status code	Description	schema
200	OK	restful api response object«Room info dto»
201	Created	
401	Unauthorized	
403	Forbidden	
404	Not Found	

Response Parameter:

Parameter	Description	Type	schema
code	response code	string	

Parameter	Description	Type	schema
data	response data	Room info dto	Room info dto
roomId	room id	integer(int64)	
roomName	room name	string	
roomType	room type	integer(int32)	
timeZone	Room time zone(UTC/GMT-5)	string	
msg	response description message	string	

Response Example:


```
{
        "code": "200",
        "data": {
                "roomId": 0,
                "roomName": "",
                "roomType": 0,
                "timeZone":  ""
                },
        "msg": "success"
}
```


3.15. To get the Room information

Url:/api/v1/room/details-info/{roomId}

Method: GET

Produces: application/x-www-form-urlencoded

Consumes: */*

Note: To get information of the Room, including the basic information like name and type, as well as a list of radars that are assigned to this Room and whether the invasion mode is ON ('arm' for the Wavve hub and 'leaveRoom" for the Assure).

Request parameter:

Parameter	Description	In	Required?	Data type	schema
Authorization	Api access token	header	true	string	
roomId	roomId	path	true	integer(int64)	

Response Status:

Status code	Description	schema
200	OK	restful api response object«Room details info dto»

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Status code	Description	schema
401	Unauthorized	
403	Forbidden	
404	Not Found	

Response Parameter:

Parameter	Description	Type	schema
code	response code	string	
data	response data	Room details info dto	Room details info dto
arm	Invasion mode， ON：true OFF：false	boolean	
assignedRadars	The list of radars that are assigned to this Room	array	string
leaveRoom	Invasion mode， ON：true OFF：false	boolean	
roomId	room id	integer(int64)	
roomName	room name	string	
roomType	room type	integer(int32)	
timeZone	Room time zone(UTC | UTC-5 | GMT-5)	string	
msg	response description message	string	

Response Example:


```
{
        "code": "200",
        "data": {
                "arm": false,
                "assignedRadars": [],
                "leaveRoom": false,
                "roomId": 0,
                "roomName": "",
                "roomType": 0,
```

```
                    "timeZone":""
                    },
          "msg": "success"
}
```


3.16. To update room information

Url:/api/v1/room/update

Method:POST

Produces:application/x-www-form-urlencoded,application/json

Consumes:*/*

Note: to update the room information according to the room id.

Example:


```
{
 "roomId": 0,
 "roomName": "",
"timeZone": ""
}
```


Request parameter:

Parameter	Description	In	Required?	Data type	schema
Authorization	Api access token	header	true	string	
room update dto	Room update dto	body	true	Room update dto	Room update dto
roomId	roomId		true	integer(int64)	
roomName	roomName		false	string	
timeZone	Room time zone (UTC | UTC-5 | GMT-5)		false	string	

Response Status:

Status code	Description	schema
200	OK	restful api response object«Void»
201	Created	
401	Unauthorized	
403	Forbidden	
404	Not Found	

Response Parameter:

Parameter	Description	Type	schema
code	response code	string	
msg	response description message	string	

Response Example:

```
{
        "code": "200",
        "msg": "success"
}
```

3.17. To set the invasion mode of the Room

Url:/api/v1/room/set-working-mode

Method: POST

Produces: application/x-www-form-urlencoded, application/json

Consumes: */*

Note: to set the invasion mode of the Room, ‘arm’ for the Wavve hub and ‘leaveRoom’ for the Assure. When the room is in invasion mode, an invasion alert will be triggered if a moving object is detected within the FOV of the radar. If it is void, no edit will be made. If the radar is re-assigned to another room, the settings of this function will be automatically changed to the same as the new room.

Example:

```
{
 "arm": false,
 "leaveRoom": false,
 "roomId": 0
}
```

Request parameter:

Parameter	Description	In	Required?	Data type	schema
Authorization	Api access token	header	true	string	
room working mode dto	Room working mode dto	body	true	Room working mode dto	Room working mode dto
arm	Invasion mode， ON：true		false	boolean	

AEROSENSE

Parameter	Description	In	Required?	Data type	schema
	OFF ： false				
leaveRoom	Invasion mode ， ON ： true OFF ： false		false	boolean	
roomId	roomId		true	integer(int64)	

Response Status:

Status code	Description	schema
200	OK	restful api response object«Void»
201	Created	
401	Unauthorized	
403	Forbidden	
404	Not Found	

Response Parameter:

Parameter	Description	Type	schema
code	response code	string	
msg	response description message	string	

Response Example:


```
{
        "code": "200",
        "msg": "success"
}
```


3.18. To edit the Room Settings

Url:/api/v1/room-set/update

Method: POST

Produces: application/x-www-form-urlencoded, application/json

Consumes: */*

Note: to edit settings of all the functions that are associated with the Room. If any of the settings fails, the corresponding function ID will be returned.

Set value range description(Please refer to the Alert Type in （Chapter 2.3） to set the functionAlert and functionRemind. It requires different description for different function. The functionRemind will trigger to push Remind, and functionAlert to push Alert.):

If fucntionId = 1, input minutes, value range 60-360, step size 30;

If functionId = 2 or 13 , input times，value range 1-10;

If functionId = 3, input minutes, value range 30-720, step size 30;
If functionId = 4 , input minutes, min = 21:00 -> 21*60 =1260 max = 01:00 -> 1440+60 = 1500, step size 30;
If functionId = 7, input minutes, value range 60-1440, step size 60;
If functionId = 8 or 9, input minutes, value range 60-720, step size 30;
If functionId = 12, input minutes, value range 5-120, step size 5;

Example:

```
{
 "radarId": "",
 "roomFunctionSetList": [
   {
     "functionAlert": 0,
     "functionId": 0,
     "functionRemind": 0,
     "functionSwitch": 0
   }
 ]
}
```

Request parameter:

Parameter	Description	In	Required?	Data type	schema
Authorization	Api access token	header	true	string	
update room function set	Update room function setting	body	true	Update room function set	Update room function set
radarId	radar id		true	string	
roomFunctionSetList	function setting list		false	array	function set
functionAlert	function alert value		false	integer	
functionId	1: Oversleeping Alert; 2: Frequent Night Activity; 3: Wandering Alert (Night);		false	integer	

Parameter	Description	In	Required?	Data type	schema
	4: Abnormal Sleep; 7: Wandering Alert (day); 8: Continuous Sedentary Notification; 9: Daily Accumulative Sedentary Notification; 11: Nutrition Alert; 12: Extended toileting; 13: Frequent Toileting (Night).				
functionRemind	function remind value		false	integer	
functionSwitch	function switch only 0 or 1, 0 off 1 on		false	integer	

Response Status:

Status code	Description	schema
200	OK	restful api response object«Room function set update response»
201	Created	
401	Unauthorized	
403	Forbidden	
404	Not Found	

Response Parameter:

Parameter	Description	Type	schema
code	response code	string	
data	response data	Room function set update response	Room function set update response
updateFailFunctions	List of failed function ID.	array	integer
msg	response description message	string	

Response Example:

```
{
        "code": "200",
        "data": {
                "updateFailFunctions": []
        },
        "msg": "success"
}
```


3.19. To get the Room settings of the radar

Url:/api/v1/room-set/list

Method: POST

Produces: application/x-www-form-urlencoded, application/json

Consumes: */*

Note: To enter a function ID to get all the information of this specific function. If it is NULL, all the information of all the functions will be returned.

Example:


```
{
 "functionIdList": [],
 "radarId": ""
}
```


Request parameter:

Parameter	Description	In	Required?	Data type	schema
Authorization	Api access token	header	true	string	
get room function set	Get room function setting	body	true	Get room function set	Get room function set
functionIdList	1: Oversleeping Alert; 2: Frequent Night Activity; 3: Wandering Alert (Night); 4: Abnormal Sleep; 7: Wandering Alert (day); 8: Continuous Sedentary Notification; 9: Daily Accumulative Sedentary Notification; 11: Nutrition Alert; 12: Extended toileting; 13: Frequent Toileting		false	array	integer

Parameter	Description	In	Required?	Data type	schema
	(Night). If function type is null, get all function				
radarId	radar id		true	string	

Response Status:

Status code	Description	schema
200	OK	restful api response object«Room function set list»
201	Created	
401	Unauthorized	
403	Forbidden	
404	Not Found	

Response Parameter:

Parameter	Description	Type	schema
code	response code	string	
data	response data	Room function set list	Room function set list
radarId	radar id	string	
roomFunctionSetList	function list	array	Room function set
functionAlert	function alert value	integer	
functionId	function id	integer	
functionRemind	function remind value	integer	
functionSwitch	function switch	integer	
roomType	radar bound room type	integer(int32)	
msg	response description message	string	

Response Example:


```
{
        "code": "200",
        "data": {
                "radarId": "",
                "roomFunctionSetList": [
```

```
                                        {
                                                    "functionAlert": 0,
                                                    "functionId": 0,
                                                    "functionRemind": 0,
                                                    "functionSwitch": 0,
                                        }
                            ],
                            "roomType": 0
                },
                "msg": "success"
}
```


3.20. To get the daily report of a single radar (one user with only one radar)

Url:/api/v1/daily/report/{radarId}/{day}

Method: GET

Produces: application/x-www-form-urlencoded

Consumes: */*

Note: To get the daily report of a radar on any given date, for example the daily report on 20230323 (in the format of yyyyMMdd), via radarID and the date. When the enduser clicks on his software or APP to request the daily report, the request will be processed via this interface and the data will be rendered by your server and presented to the enduser. Please note that this interface only applies to the scenario that there is only one radar registered under a user. For one user with multiple radars, please refer to 3.21.

Please note that the Room the radar is assigned to must be in the right time zone as where the Room is physically located. Otherwise, the report may not be accurate. The user can edit the time zone of the Room using the information of the Room Settings.

Please note that if the requested date is the day before and the enduser is still in bed or it's less than an hour after the enduser gets up, the sleep event is still in progress, or the sleep report is not completed yet. Therefore there will be no sleep data in the report generated via this interface. To get a complete report, please request this interface after the user has been up for more than an hour.

Please note that our cloud only supports requests of daily reports up to one month ago. Otherwise the returned data will be ERROR.

Request parameter:

Parameter	Description	In	Required?	Data type	schema

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Parameter	Description	In	Required?	Data type	schema
Authorization	Api access token	header	true	string	
radarId	radarId	path	true	string	
Date	yyyyMMdd	path	true	string	

Response Status:

Status code	Description	schema
200	OK	restful api response object«daily report response dto»
401	Unauthorized	
403	Forbidden	
404	Not Found	

Response Parameter:

Parameter	Description	Type	schema
code	response code	string	
data	response data	daily report response dto	daily report response dto
dailyReports	real daily report data	array	DailyReport
reportEndTime	Report end time(yyyy-MM-dd HH:mm:ss)	string(date-time)	
reportStartTime	Report start time(yyyy-MM-dd HH:mm:ss)	string(date-time)	
reportZonedEndTime	Report zoned end time(ISO-8601)	string(date-time)	
reportZonedStartTime	Report zoned start time(ISO-8601)	string(date-time)	
reportType	Report typeCommon Enumeration Type	integer	
todayNum	Report date. Format:yyyyMMdd	integer	
value	Report value	number	
radarId	radar	string	
msg	response description message	string	

Response Example:


```
{
        "code": "200",
        "data": {
                "dailyReports": [
                        {
                                "reportEndTime": "",
                                "reportStartTime": "",
                                    "reportZonedEndTime":"",
                                    "reportZonedStartTime":"",
                                "reportType": 0,
                                "todayNum": 0,
                                "value": 0
                        }
                ],
                "radarId": ""
        },
        "msg": "success"
}
```


3.21. To obtain the daily report via the radarID (one user with multiple radars)

Url:/api/v1/daily/report/by-radar-list

Method:POST

Produces:application/x-www-form-urlencoded,application/json

Consumes:*/*

Note: If there are several radars registered under the same user, the developer can obtain the daily report generated by each radar on any given date via the radarID. Please note that although one user can have multiple Assures, there can be only one Wavve registered with the same user. Please refer to example codes below for daily report of a radarID:

Example:


```
{
 "date": "",
 "radarIds": []
}
```


Request parameter:

Parameter	Description	In	Required?	Data type	schema
Authorization	Api access token	header	true	string	
dailyReportListReqDTO	GetDailyReportByRadarListDTO	body	true	GetDailyReportByRadarListDTO	GetDailyReportByRadarListDTO

Parameter	Description	In	Required?	Data type	schema
date	The report date format：yyMMdd		true	string	
radarIds	Radar id list		true	array	string

Response Status:

Status code	Description	schema
200	OK	restful api response object«List«Daily report response dto»»
201	Created	
401	Unauthorized	
403	Forbidden	
404	Not Found	

Response Parameter:

Parameter	Description	Type	schema
code	response code	string	
data	response data	array	Daily report response dto
dailyReports	Real daily report data	array	Daily report data
reportEndTime	Report end time(yyyy-MM-dd HH:mm:ss)	string(date-time)	
reportStartTime	Report start time(yyyy-MM-dd HH:mm:ss)	string(date-time)	
reportZonedEndTime	Report zoned end time(ISO-8601)	string(date-time)	
reportZonedStartTime	Report zoned start time(ISO-8601)	string(date-time)	
reportType	Report type	integer	
todayNum	Report date. format: yyyyMMdd	integer	
value	Report value	number	
radarId	radarId	string	
msg	response description message	string	

Response Example:

```
{
        "code": "200",
        "data": [
                {
                        "dailyReports": [
                                {
                                        "reportEndTime": "",
                                        "reportStartTime": "",
                                           "reportZonedEndTime":"",
                                           "reportZonedStartTime":"",
                                        "reportType": 0,
                                        "todayNum": 0,
                                        "value": 0
                                }
                        ],
                        "radarId": ""
                }
        ],
        "msg": "success"
}
```


3.22. To turn off and leave radar zigbee gateway (Wavve hub)

Url:/api/v1/zigbee-network/close/{radarId}

Method:POST

Produces:application/x-www-form-urlencoded

Consumes:*/*

Note:

- When this api is executed, the Zigbee network of the hub will be completed shut done is no specific deviceMac of a sub-device is assigned, and all the connection to all the sub-devices will be turned off. It means the Zigbee hub is reset. In this case, you will need to establish the Zigbee network of the hub again as described in Section 3.23, and all the sub-devices will be re-paired to connect to the hub.   .
- When this api is executed and the deviceMac of a specific sub-device is assigned, only the Zigbee of this sub-device will be turned off and the offline message will be pushed to the server. Other sub-devices will not be affected..
- Please note that if the **removeDevice** flag is set as **true**, all the information of that sub device will be deleted from the database. However, it will not be deleted if it fails to turn off the Zigbee network of the sub-device.

Example:


```
{
  “deviceMac": “”,
“removeDevice”:false
}
```

Request parameter:

Parameter	Description	In	Required?	Data type	schema
Authorization	Api access token	header	true	string	
radarId	radarId	path	true	string	
closeZigbeeNetworkDTO	CloseZigbeeNetworkDTO	body	false	CloseZigbeeNetworkDTO	CloseZigbeeNetworkDTO
deviceMac	deviceMac		false	string	
removeDevice	removeDevice		false	String	

Response Status:

Status code	Description	schema
200	OK	restful api response object«Void»
201	Created	
401	Unauthorized	
403	Forbidden	
404	Not Found	

Response Parameter:

Parameter	Description	Type	schema
code	response code	string	
msg	response description message	string	

Response Example:

```
{
        "code": "200",
        "msg": "success"
}
```

3.23. To establish the zigbee network of the hub and to add sub-device to the network (Wavve hub)

Url:/api/v1/zigbee-network/open/{radarId}

Method:POST

Produces:application/x-www-form-urlencoded,application/json

Consumes:*/*

Note:

- To enable the zigbee gateway of the hub,and allow the sub-devices to connect to the hub within the designated time frame set by the **waitJoinNetSeconds** parameter.
- To set **Waiting Time** (waitJoinNetSeconds)
  - If set at 180: once the network is enabled, it will scan continuously for 180 seconds. During this time, the sub-devices can be connectedto the hub. After 180s, the network of the hub will be closed. The developer can re-send this commend before the end of the 180s to restart the clock and initiate a new scan, or alternatively to manually stop the current scan by sending a "waitJoinNetSeconds".
  - If set as **0, the connection will be immediately stopped and no new device can be connected. The developer can also manually close the zigbee network during the waiting period**.
- Once the add-on is successfully connected to the hub, it will automatically report for registration and pair to the radar hub. The connection information of the add-ons will then be pushed to your service.
- Please save the message to respond to request to check the connection status of the add-ons, or to request the connection information of the add-ons through the check-add-on-list interface.
- For devices with communication firmware of version 02.04.21.33 and above: the device will continuously blink red during the Zigbee network scanning until the scanning process is completed.

Example:

```
{
 "waitJoinNetSeconds": 0
}
```

Request parameter:

Parameter	Description	In	Required?	Data type	schema

Parameter	Description	In	Required?	Data type	schema
Authorization	Api access token	header	true	string	
radarId	radarId	path	true	string	
openZigbeeNetworkDTO	OpenZigbeeNetworkDTO	body	true	OpenZigbeeNetworkDTO	OpenZigbeeNetworkDTO
waitJoinNetSeconds	allowJoinNetSeconds		false	integer(int32)	

Response Status:

Status code	Description	schema
200	OK	restful api response object«Void»
201	Created	
401	Unauthorized	
403	Forbidden	
404	Not Found	

Response Parameter:

Parameter	Description	Type	schema
code	response code	string	
msg	response description message	string	

Response Example:

```
{
        "code": "200",
        "msg": "success"
}
```

3.24. To request information of the add-ons that are paired to the radar hub (Wavve hub)

Url:/api/v1/sub-device/list/{radarId}

Method:POST

Produces:application/x-www-form-urlencoded

Consumes:*/*

Note: Once the add-on is connected via the radar it will be automatically paired to the hub and to the room that the radar is assigned to.

Request parameter:

Parameter	Description	In	Required?	Data type	schema
Authorization	Api access token	header	true	string	
radarId	radarId	path	true	string	

Response Status:

Status code	Description	schema
200	OK	restful api response object«List«SubDeviceDTO»»
201	Created	
401	Unauthorized	
403	Forbidden	
404	Not Found	

Response Parameter:

Parameter	Description	Type	schema
code	response code	string	
data	response data	array	SubDeviceDTO
deviceMac	Sub device MAC address	string	
deviceName	Sub device name	string	
deviceType	Sub device type. Please refer to the enumeration value list for specific values	integer(int32)	
onlineState	Sub device online state. true: online false: offline	boolean	
radarId	radar id	string	
roomId	Assigned room id	integer(int64)	
msg	response description message	string	

Response Example:


```
{
        "code": "200",
        "data": [
                {
```

```
                              "deviceMac": "",
                              "deviceName": "",
                              "deviceType": 0,
                              "onlineState": false,
                              "radarId": "",
                              "roomId": 0
                    }
          ],
          "msg": "success"
}
```


3.25. To edit information of the add-ons that are paired to the radar hub (Wavve hub)

Url:/api/v1/sub-device/update/{radarId}

Method:POST

Produces:application/x-www-form-urlencoded,application/json

Consumes:*/*

Note: To edit information of the add-ons that are paired to the radar. No modification will be made if nothing is updated.

Example:


```
{
 "deviceMac": "",
 "deviceName": ""
}
```


Request parameter:

Parameter	Description	In	Required?	Data type	schema
Authorization	Api access token	header	true	string	
radarId	radarId	path	true	string	
updateSubDeviceDTO	UpdateSubDeviceDTO	body	true	UpdateSubDeviceDTO	UpdateSubDeviceDTO
deviceMac	Sub device MAC address		true	string	
deviceName	Sub device name		false	string	

Response Status:

Status code	Description	schema

Status code	Description	schema
200	OK	restful api response object«Void»
201	Created	
401	Unauthorized	
403	Forbidden	
404	Not Found	

Response Parameter:

Parameter	Description	Type	schema
code	response code	string	
msg	response description message	string	

Response Example:


```
{
	"code": "200",
	"msg": "success"
}
```


3.26. Delete a Sub-Device that is bound to the Hub (Wavve Hub)

Url:/api/v1/sub-device/delete/{radarId}/{deviceMac}

Method:DELETE

Produces:application/x-www-form-urlencoded,application/json

Consumes:*/*

Note:

- You will not get the offline notice by directly deleting the sub-device from the database of the Hub that it is bound to, not matter whether device is online.
- This interface is suitable for removing the data of any sub-device from the database when the sub-device is already offline.

Example:

Request parameter:

Parameter	Description	In	Required?	Data type	schema
Authorization	Api access token	header	true	string	

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Parameter	Description	In	Required?	Data type	schema
radarId	radarId	path	true	string	
deviceMac	Sub device MAC address	path	true	string	

Response Status:

Status code	Description	schema
200	OK	restful api response object«Void»
201	Created	
401	Unauthorized	
403	Forbidden	
404	Not Found	

Response Parameter:

Parameter	Description	Type	schema
code	response code	string	
msg	response description message	string	

Response Example:


```
{
        "code": "200",
        "msg": "success"
}
```


3.27. To request the ARM/DISARM timer of the room that the radar is assigned to (Wavve hub)

Url:/api/v1/room-arm-schedule/list/{radarId}

Method:GET

Produces:application/x-www-form-urlencoded

Consumes:*/*

Note:

Request parameter:

Parameter	Description	In	Required?	Data type	schema

AEROSENSE

Parameter	Description	In	Required?	Data type	schema
Authorization	Api access token	header	true	string	
radarId	radarId	path	true	string	

Response Status:

Status code	Description	schema
200	OK	restful api response object«List«ArmScheduleSetDTO»»
401	Unauthorized	
403	Forbidden	
404	Not Found	

Response Parameter:

Parameter	Description	Type	schema
code	response code	string	
data	response data	array	ArmScheduleSetDTO
endTime	Disarm time. If null, the system will be on ARM constantly. The startTime and endTime cannot be both null. Time format: ISO-8601	string(date-time)	
id	Setup the id	integer(int64)	
scheduleState	ON/FF setting: 0 (OFF); 1 (ON)	integer(int32)	
scheduleType	Timer type: \n Once: just this once \n: on a daily base \n 1, 2, 3, 4, 5, 6, 7: customizable. 1=Monday, 2=Tuesday, 3=Wednesday, 4=Thursday, 5=Friday, 6=Saturday, 7=Sunday.	string	
startTime	Arm time. startTime and endTime cannot be both null. Time format: ISO-8601	string(date-time)	
msg	response description message	string	

Response Example:

```
{
        "code": "200",
        "data": [
                {
                        "endTime": "",
                        "id": 0,
                        "scheduleState": 0,
                        "scheduleType": "",
                        "startTime": ""
                }
        ],
        "msg": "success"
}
```


3.28. To setup a new ARM/DISARM timer of the room that the radar is assigned to (Wavve hub)

Url:/api/v1/room-arm-schedule/add/{radarId}

Method:POST

Produces:application/x-www-form-urlencoded,application/json

Consumes:*/*

Note: To setup a new timer of the room that the radar is assigned to. If the timer is a one-time setting, it can be triggered at a certain time on a certain day in the future. Otherwise it has to be a certain time but without the date. Please note:

Please note that it is requested that the radar is assigned to a room that has the correct time zone information before the radar is configured. The time zone of the room must the same as the set time zone. If the time zone of the room is revised, so will be the set time.

1. The settings here are paired to the room that the device is assigned to, and will not be affected if the radar or the add-ons within the room is changed.
2. If the time of the timer that you set is earlier than the current time, it won't be effective until the next day. For example, if you set up a timer of 8 am at 9 am, the timer will be triggered at 8am the next day.

Example:


```
{
 "endTime": "",
 "scheduleType": "",
 "startTime": ""
}
```


Request parameter:

Parameter	Description	In	Requir ed?	Data type	schema

Parameter	Description	In	Required?	Data type	schema
Authorization	Api access token	header	true	string	
radarId	radarId	path	true	string	
addArmScheduleSetDTO	AddArmScheduleSetDTO	body	true	AddArmScheduleSetDTO	AddArmScheduleSetDTO
endTime	Disarm time. If null, the system will be on ARM constantly. The startTime and endTime cannot be both null. Time format: ISO-8601		false	string(date-time)	
scheduleType	Timer type: \n Once: just this once \n: on a daily base \n 1, 2, 3, 4, 5, 6, 7: customizable. 1=Monday, 2=Tuesday, 3=Wednesday, 4=Thursday, 5=Friday, 6=Saturday, 7=Sunday.		false	string	
startTime	Arm time. startTime and endTime cannot be both null. Time format: ISO-8601		false	string(date-time)	

Response Status:

Status code	Description	schema
200	OK	restful api response object«Void»

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Status code	Description	schema
201	Created	
401	Unauthorized	
403	Forbidden	
404	Not Found	

Response Parameter:

Parameter	Description	Type	schema
code	response code	string	
msg	response description message	string	

Response Example:

```
{
        "code": "200",
        "msg": "success"
}
```

3.29. To delete the ARM/DISARM timer of the room that the radar is assigned to (Wavve hub)

Url:/api/v1/room-arm-schedule/{radarId}/{id}

Method:DELETE

Produces:application/x-www-form-urlencoded

Consumes:*/*

Note:

Request parameter:

Parameter	Description	In	Required?	Data type	schema
Authorization	Api access token	header	true	string	
radarId	radarId	path	true	string	
id	id	path	true	integer(int64)	

Response Status:

Status code	Description	schema
200	OK	restful api response object«Void»

Status code	Description	schema
204	No Content	
401	Unauthorized	
403	Forbidden	

Response Parameter:

Parameter	Description	Type	schema
code	response code	string	
msg	response description message	string	

Response Example:

```
{
        "code": "200",
        "msg": "success"
}
```

3.30. To edit the ON/OFF of the ARM/DISARM timer of the room that the radar is assigned to (Wavve hub)

Url:/api/v1/room-arm-schedule/set-switch

Method:POST

Produces:application/x-www-form-urlencoded,application/json

Consumes:*/*

Note:

Example:

```
{
  "id": 0,
  "radarId": "",
  "scheduleState": 0
}
```

Request parameter:

Parameter	Description	In	Required?	Data type	schema
Authorization	Api access token	header	true	string	
setRoomArmSche	To edit	bod	true	SetRoomArmSchedul	SetRoomArmSchedul

Parameter	Description	In	Required?	Data type	schema
duleSwitch	the ON/OFF of the ARM/DISARM timer of the room that the radar is assigned to.	y		eSwitchDTO	eSwitchDTO
id	set id		true	integer(int64)	
radarId	radar id		true	string	
scheduleState	schedule state		true	integer(int32)	

Response Status:

Status code	Description	schema
200	OK	restful api response object«Void»
201	Created	
401	Unauthorized	
403	Forbidden	
404	Not Found	

Response Parameter:

Parameter	Description	Type	schema
code	response code	string	
msg	response description message	string	

Response Example:


```
{
        "code": "200",
        "msg": "success"
}
```


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3.31. To request the DELAY SETTING of the room that the radar is assigned to (Wavve hub)

Url:/api/v1/room-delay-alarm/{radarId}

Method:GET

Produces:application/x-www-form-urlencoded

Consumes:*/*

Note: The DELAY SETTING includes entry-delay-duration and exit-delay-duration. The entry_delay_ duration is the time interval between trigger an alert by entering the room and sending out the alert. The alert that is triggered will only be sent after this time interval. For example, if the entry-delay-duration is set at 30s, and the system detects an entry at 7pm, the alert will be sent at 7:00:30pm. If the user manually deletes the arm during the time frame, the alert will not be sent. The exit_delay_duration is the time interval between the set ARM time and the system will actually send an alert. For example, if the ARM time is set at 8:00am and the exit-delay-duration is set at 30s, and user actually leaves the room at 8:00:25am, the system will be mute and no alert will be sent to the user.

Request parameter:

Parameter	Description	In	Required?	Data type	schema
Authorization	Api access token	header	true	string	
radarId	radarId	path	true	string	

Response Status:

Status code	Description	schema
200	OK	restful api response object«DelayAlarmSetDTO»
401	Unauthorized	
403	Forbidden	
404	Not Found	

Response Parameter:

Parameter	Description	Type	schema
code	response code	string	
data	response data	DelayAlarmSetDTO	DelayAlarmSetDTO

Parameter	Description	Type	schema
entryDelayDuration	entry_delay_duration: the time interval between trigger an alert by entering the room and sending out the alert. It has to be between 0-60 seconds. The alert that is triggered will only be sent after this time interval. If the user manually deletes the arm during the time frame, the alert will not be sent. 0 means the function is off.	integer(int32)	
exitDelayDuration	exit_delay_duration: the time interval between the set ARM time and the system will actually send an alert. It has to be between 0-60 seconds. For example, if the ARM time is set at 8:00am and the exit-delay-duration is set at 30s, and user actually leaves the room at 8:00:25, the system will be mute and no alert will be sent to the user. 0 means the function is off.	integer(int32)	
id	Setup id	integer(int64)	
msg	response description message	string	

Response Example:


```
{
    "code": "200",
    "data": {
        "entryDelayDuration": 0,
        "exitDelayDuration": 0,
        "id": 0
    },
    "msg": "success"
}
```


3.32. To setup a new DELAY SETTING of the room that the radar is assigned to (Wavve hub)

Url:/api/v1/room-delay-alarm/add/{radarId}

Method:POST

Produces:application/x-www-form-urlencoded,application/json

Consumes:*/*

Note: The DELAY SETTING includes entry-delay-duration and exit-delay-duration. The entry_delay_ duration is the time interval between trigger an alert by entering the room and sending out the alert. The alert that is triggered will only be sent after this time interval. For example, if the entry-delay-duration is set at 30s, and the system detects an entry at 7pm, the alert will be sent at 7:00:30pm. If the user manually deletes the arm during the time frame, the alert will not be sent. The exit_delay_duration is the time interval between the set ARM time and the system will actually send an alert. For example, if the ARM time is set at 8:00am and the exit-delay-duration is set at 30s, and user actually leaves the room at 8:00:25am, the system will be mute and no alert will be sent to the user.

Please note that these settings are paired to the Room. If the devices within the room are changes, the settings will still be the same.

Example:

```
{
 "entryDelayDuration": 0,
 "exitDelayDuration": 0
}
```

Request parameter:

Parameter	Description	In	Required?	Data type	schema
Authorization	Api access token	header	true	string	
radarId	radarId	path	true	string	
addDelayAlarmSetDTO	AddDelayAlarmSetDTO	body	true	AddDelayAlarmSetDTO	AddDelayAlarmSetDTO
entryDelayDuration	entry_delay_duration: the time interval between trigger an alert by entering the room and sending out the alert. It has to be between 0-60 seconds. The alert that is triggered will only be sent after this time interval. If the user manually deletes the arm during the time frame, the alert will not be sent. 0 means the function is off.		true	integer(int32)	
exitDelayDurat	exit_delay_duration: the time interval between the set ARM time		true	integer(int3	

Parameter	Description	In	Required?	Data type	schema
ion	and the system will actually send an alert. It has to be between 0-60 seconds. For example, if the ARM time is set at 8:00am and the exit-delay-duration is set at 30s, and user actually leaves the room at 8:00:25, the system will be mute and no alert will be sent to the user. 0 means the function is off.			2)	

Response Status:

Status code	Description	schema
200	OK	restful api response object«Void»
201	Created	
401	Unauthorized	
403	Forbidden	
404	Not Found	

Response Parameter:

Parameter	Description	Type	schema
code	response code	string	
msg	response description message	string	

Response Example:


```
{
        "code": "200",
        "msg": "success"
}
```


3.33. To edit the DELAY SETTING of the room that the radar is assigned to (Wavve hub)

Url:/api/v1/room-delay-alarm/update/{radarId}

Method:POST

Produces:application/x-www-form-urlencoded,application/json

Consumes:*/*

Note: The user can edit the DELAY SETTING one by one. If 0 are entered for all the DELAY SETTING, it means all the DELAY SETTING functions are off.

Example:


```
{
 "entryDelayDuration": 0,
 "exitDelayDuration": 0
}
```


Request parameter:

Parameter	Description	In	Required?	Data type	schema
Authorization	Api access token	header	true	string	
radarId	radarId	path	true	string	
updateDelayAlarmSetDTO	To edit the DELAY SETTING of the room. It can be modified individually.	body	true	UpdateDelayAlarmSetDTO	UpdateDelayAlarmSetDTO
entryDelayDuration	Entry-delay-duration. It has to be between 0-60s. 0 means the function is off.		false	integer(int32)	
exitDelayDuration	Exit-delay-duration. It has to be between 0-60s. 0 means the function is off.		false	integer(int32)	

Response Status:

Status code	Description	schema
200	OK	restful api response object«Void»
201	Created	
401	Unauthorized	
403	Forbidden	
404	Not Found	

Response Parameter:

Parameter	Description	Type	schema
code	response code	string	
msg	response description message	string	

Response Example:

```
{
        "code": "200",
        "msg": "success"
}
```

3.34. To request information of all the sirens that are in the room the radar is assigned to (Wavve hub)

Url:/api/v1/room-siren-set/list/{radarId}

Method:GET

Produces:application/x-www-form-urlencoded

Consumes:*/*

Note: The siren will be in default setting once is it connected, with low volume, no light and with beep. It will be paired to the same room as the radar. If the radar is reassigned to a different room, the siren will be automatically set to default once it's connected back to the radar hub.

Request parameter:

Parameter	Description	In	Required?	Data type	schema
Authorization	Api access token	header	true	string	
radarId	radarId	path	true	string	

Response Status:

Status code	Description	schema
200	OK	restful api response object«List«SirenSetDTO»»
201	Created	
401	Unauthorized	
403	Forbidden	
404	Not Found	

Response Parameter:

Parameter	Description	Type	schema
code	response code	string	
data	response data	array	SirenSetDTO
alertTime	The length of the alert. Unit: s. It must be	integer(int32)	

Parameter	Description	Type	schema
	between 1-300 with a default value of 5s.		
armSwitch	Beep switch: false: OFF; true: ON	boolean	
deviceMac	Sub device MAC address	string	
lightSwitch	Light switch: false: OFF; true: ON	boolean	
quietModeSwitch	Quiet-mode switch: false: OFF; true: ON	boolean	
volume	Volume: 0 ：low; 1 ：medium; 2 ：high.	integer(int32)	
msg	response description message	string	

Response Example:


```
{
    "code": "200",
    "data": [
        {
            "alertTime": 0,
            "armSwitch": false,
            "deviceMac": "",
            "lightSwitch": false,
            "quietModeSwitch": false,
            "volume": 0
        }
    ],
    "msg": "success"
}
```


3.35. To edit the settings of a specific siren that is in the room the radar is assigned to (Wavve hub)

Url:/api/v1/room-siren-set/update/{radarId}

Method:POST

Produces:application/x-www-form-urlencoded,application/json

Consumes:*/*

Note: To edit the settings of the siren by editing the corresponding parameters (silence mode, light, volume and beep switches). No modification will be made if the parameters are not changed.

Example:


```
{
 "alertTime": 0,
 "armSwitch": false,
 "deviceMac": "",
 "lightSwitch": false,
 "quietModeSwitch": false,
 "volume": 0
}
```


Request parameter:

Parameter	Description	In	Required?	Data type	schema
Authorization	Api access token	header	true	string	
radarId	radarId	path	true	string	
sirenSetDTO	SirenSetDTO	body	true	SirenSetDTO	SirenSetDTO
alertTime	The length of the alert. Unit: s. It must be between 1-300 with a default value of 5s.		false	integer(int32)	
armSwitch	Beep switch: false: OFF; true: ON		false	boolean	
deviceMac	Sub device MAC address		true	string	
lightSwitch	Light switch: false: OFF; true: ON		false	boolean	
quietModeSwitch	Quiet-mode switch: false: OFF; true: ON		false	boolean	
volume	Volume: 0：low; 1：medium; 2：high.		false	integer(int32)	

Response Status:

Status code	Description	schema
200	OK	restful api response object«Void»
201	Created	
401	Unauthorized	
403	Forbidden	
404	Not Found	

Response Parameter:

Parameter	Description	Type	schema
code	response code	string	
msg	response description message	string	

Response Example:

```
{
        "code": "200",
        "msg": "success"
}
```

3.36. To check if the radar can be updated

url:/api/v1/firmware/upgrade-info/{radarId}

method:GET

produces:application/x-www-form-urlencoded

consumes:*/*

Note: To check if the radar can be updated. If so, you will be responded with the firmware version, firmware description and whether the update is in progress.

Params:

name	description	in	require	type	schema
Authorization	Api access token	header	true	string	
radarId	radarId	path	true	string	

Status:

code	description	schema
200	OK	restful api response object«RadarFirmwareUpgradeInfoDTO»

code	description	schema
401	Unauthorized	
403	Forbidden	
404	Not Found	

Response Params:

name	description	type	schema
code	response code	string	
data	response data	RadarFirmwareUpgradeInfoDTO	RadarFirmwareUpgradeInfoDTO
firmwareDesc	Description of the radar firmware	string	
radarVersion	Current radar firmware version	string	
upgradeable	Whether the radar can be updated to the latest firmware	boolean	
upgradeableVersion	The latest firmware the radar can ben updated to	string	
upgrading	Whether the radar will be updated within the next 24hrs.	boolean	
msg	response description message	string	

Response Example:


```
{
	"code": "200",
	"data": {
		"firmwareDesc": "",
		"radarVersion": "",
		"upgradeable": false,
		"upgradeableVersion": "",
		"upgrading": false
	},
	"msg": "success"
}
```


3.37. To upgrade the radar firmware

url:/api/v1/firmware/do-upgrade/{radarId}

method:POST

produces:application/x-www-form-urlencoded

consumes:*/*

Note: If the user selects to upgrade the radar firmware, the upgrade will be attempted within the next 24hrs. The user will receive a push message with the upgrade result within the 24hrs, whether the upgrade is successful or fails, and the upgrade process is completed. Please note that when the user requests to update the firmware, the response is immediate. However, the update results will not be returned immediately. Please obtain the update result via the push message, or to double check the interface to request for the upgrade results.

Params:

name	description	in	require	type	schema
Authorization	Api access token	header	true	string	
radarId	radarId	path	true	string	

Status:

code	description	schema
200	OK	restful api response object«FirmwareUpgradeStatusDTO»
201	Created	
401	Unauthorized	
403	Forbidden	
404	Not Found	

Response Params:

name	description	type	schema
code	response code	string	
data	response data	FirmwareUpgradeStatusDTO	FirmwareUpgradeStatusDTO
radarId	Radar ID	string	
radarVersion	Current radar firmware version	string	
upgradeResult	The upgrade result	boolean	
upgrading	Whether the radar will be updated within the next 24hrs.	boolean	

name	description	type	schema
msg	response description message	string	

Response Example:


```
{
    "code": "200",
    "data": {
        "radarId": "",
        "radarVersion": "",
        "upgradeResult": false,
        "upgrading": false
    },
    "msg": "success"
}
```


3.38. To turn ON/OFF machine learning (for fall detection of the Assure)

url:/api/v1/radar/set-machine-learning-switch

method:POST

produces:application/x-www-form-urlencoded,application/json

consumes:*/*

Note: The user can choose whether to turn on or turn off machine learning for fall detection to minimize false alerts. 1: ON. 0: OFF.

Example:


```
{
 "radarId": "",
 "status": 0
}
```


Params:

name	description	in	require	type	schema
Authorization	Api access token	header	true	string	
setMachineLearningDTO	SetMachineLearningDTO	body	true	SetMachineLearningDTO	SetMachineLearningDTO
radarId	radar id		true	string	
status	0: OFF; 1: ON		true	integer(int32)	

Status:

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code	description	schema
200	OK	restful api response object«Void»
201	Created	
401	Unauthorized	
403	Forbidden	
404	Not Found	

Response Params:

name	description	type	schema
code	response code	string	
msg	response description message	string	

Response Example:

```
{
	"code": "200",
	"msg": "success"
}
```

3.39. To request the status of machine learning (Assure)

url:/api/v1/radar/machine-learning-switch/{radarId}

method:GET

produces:application/x-www-form-urlencoded

consumes:*/*

Note: to request for the status of whether the machine learning is ON/OFF. 1: ON; 0: OFF.

Params:

name	description	in	require	type	schema
Authorization	Api access token	header	true	string	
radarId	radarId	path	true	string	

Status:

code	description	schema
200	OK	restful api response object«GetMachineLearningDTO»
401	Unauthorized	

code	description	schema
403	Forbidden	
404	Not Found	

Response Params:

name	description	type	schema
code	response code	string	
data	response data	GetMachineLearningDTO	GetMachineLearningDTO
radarId	radar id	string	
status	Status: 0: OFF; 1: ON	integer(int32)	
msg	response description message	string	

Response Example:


```
{
    "code": "200",
    "data": {
        "radarId": "",
        "status": 0
    },
    "msg": "success"
}
```


3.40. To learn the interference in the room (Assure/Wavve)

url:/api/v1/radar/save-room-interference-study-time/{radarId}

method:POST

produces:application/x-www-form-urlencoded

consumes:*/*

Note:

- Environmental noise may lead to false presence detection. Please use this interface for automatic interference learning. It is recommended to do the learning as a standard step during device installation.
- Whenever the server gets a request from this interface, the server will mark its local time as the start point. The learning will take 10 min. It is very important that the environment must fulfill the descriptions below.
- For the Assure:

- Before the installation engineer leaves the room, make sure to turn on all the potential fixed interferences in the room to the max settings, For example ventilations, fans, electric heaters that rotate, fish tank with pumps.
- Leave the room and make sure there is no body or pet inside.
- There are two different ways to learn the noise. The developer can choose either.

  1) Cloud learning. Set **radarStudy** to **false, and** the Assure will automatic scan and learn the environment energy as a threshold to decide if the room is empty. The learning is done by the cloud server and not by the radar. A successful response indicates that the learning process has started. It will take about 10 minutes to complete learning. The result will interfere with the accuracy of the ADL report.
  2) Cloud and radar learning. Set **radarStudy** to **true**. Both the Assure server algorithm (same as above) and the radar firmware will start to learn the environment. Please note that it will take the radar only 1-2 minutes to complete the process. Once completed, the result will be pushed via the /api/room-interference-study-finished interface (refer to the file your-cloud-restful-api). Please note that this is only applicable to devices with firmware of **2.8.2.58 and later**.

- For the Wavve:
  - Make sure there is no body or pet around the bed.
  - Set **radarStudy** to **true**, and the Wavve will automatic scan and learn the environment energy as a threshold to decide if the bed is empty.. It will take about 30 seconds. Please note that this is only applicable to devices with firmware of 2.1.77.0 and later (if it is V2.x.x.x), or 3.0.0.25 and later (if it is V3.x.x.x).

Example:

```
{
 "radarStudy":false
}
```

Params:

name	description	in	require	type	schema
Authorization	Api access token	header	true	string	
radarId	radarId	path	true	string	
roomInterferenceSt	RoomInterferenceStud	body	false	RoomIn	RoomInterf

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name	description	in	require	type	schema
udyReqDTO	yReqDTO			terferenceStudyReqDTO	erenceStudyReqDTO
radarStudy	Radar interference learning identifier	Body	false	boolean	

Status:

code	description	schema
200	OK	restful api response object«boolean»
201	Created	
401	Unauthorized	
403	Forbidden	
404	Not Found	

Response Params:

name	description	type	schema
code	response code	string	
data	response data	boolean	
msg	response description message	string	

Response Example:


```
{
        "code": "200",
        "data": false,
        "msg": "success"
}
```


3.41. To get the environment interference threshold (Assure/Wavve)

url:/api/v1/radar/room-interference/{radarId}

method:GET

produces:application/x-www-form-urlencoded

consumes:*/*

Note:

Get the interference threshold of the current room for the Assure or the bed for the Wavve. It is applicable to Assure with firmware versions of 2.8.2.58 and later, and Wavve of 2.1.77.0 and later (or 3.0.0.25 and later).

Params:

name	description	in	require	type	schema
Authorization	Api access token	header	true	string	
radarId	radarId	path	true	string	

Status:

code	description	schema
200	OK	restful api response object«boolean»
201	Created	
401	Unauthorized	
403	Forbidden	
404	Not Found	

Response Params:

name	description	type	schema
code	response code	string	
data	response data	RoomInterferenceInfoDTO	RoomInterferenceInfoDTO
radarId	radarId	string	
threshold	Environment interference threshold	float(32)	
msg	response description message	string	

Response Example:


```
{
        "code": "200",
        "data": {
                "radarId": "",
                "threshold": 0.0
        },
        "msg": "success"
}
```


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3.42. To set room interference environment threshold (Assure/Wavve)

url:/api/v1/radar/room-interference

method:PUT

produces:application/x-www-form-urlencoded

consumes:*/*

Note:

The developer can use this interface to manually set the environment interference threshold. It is applicable to Assure with firmware versions of 2.8.2.58 and later, and Wavve of 2.1.77.0 and later (or 3.0.0.25 and later).

Example:

```
{
 "radarId":"",
   "threshold":0.0
}
```

Params:

name	description	in	require	type	schema
Authorization	Api access token	header	true	string	
radarId	radarId	path	true	string	
roomInterferenceInfoDTO	RoomInterferenceInfoDTO	body	true	RoomInterferenceInfoDTO	RoomInterferenceInfoDTO
radarId	radarId	body	true	string	
threshold	Environment interference threshold	body	true	float(32)	

Status:

code	description	schema
200	OK	restful api response object«boolean»
201	Created	
401	Unauthorized	

code	description	schema
403	Forbidden	
404	Not Found	

Response Params:

name	description	type	schema
code	response code	string	
msg	response description message	string	

Response Example:

```
{
        "code": "200",
        "msg": "success"
}
```

3.43. To setup the room layout manually (Assure)

url:/api/v1/room-configuration-area/{radarId}

method:PUT

produces:application/x-www-form-urlencoded,application/json

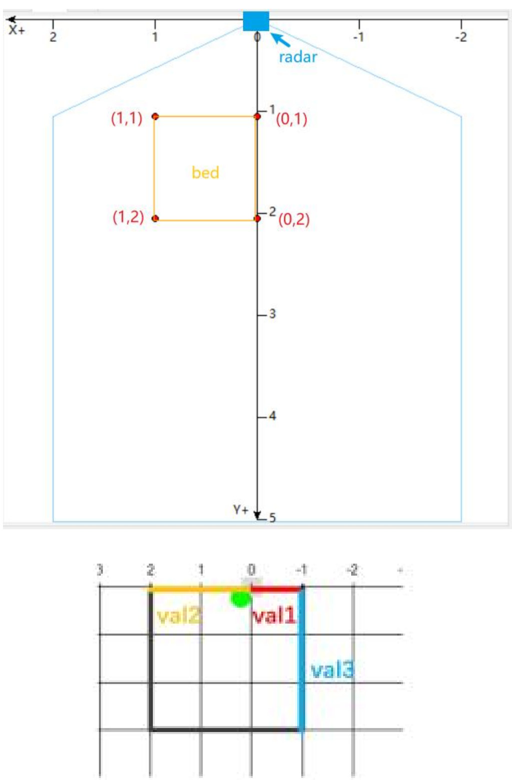
consumes:*/*

Note: To setup the coordinates that define whether a certain area within the room is a door, a bed or a sofa. There can be at most 5 such areas and each area must be in the shape of a rectangle.

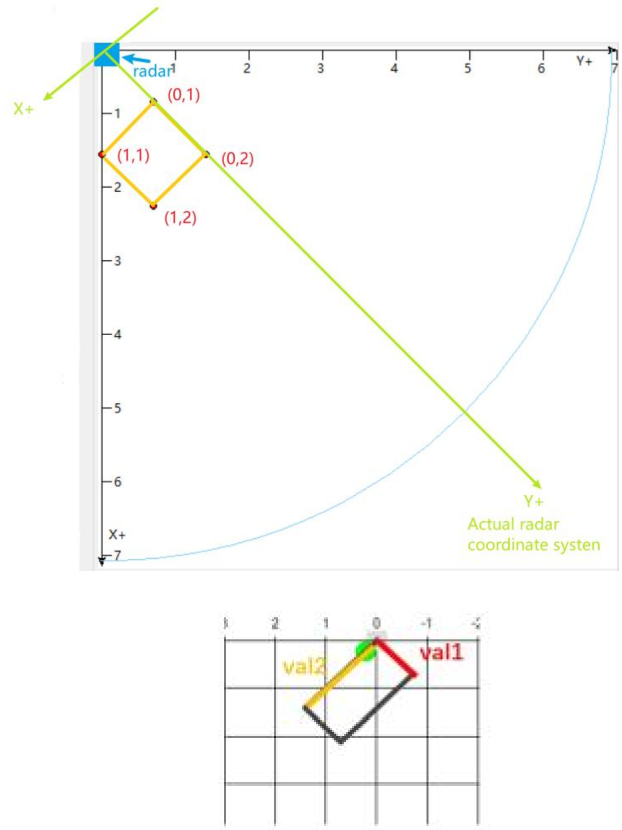
Please note that the previous settings will be covered up by the new settings.

This function is only available for devices with firmware of 2.8.x.x and later versions.

See diagram below for the definition of the coordinate system (unit: meter).



Wall mount



Corner mount

Example:

```
{
 "areaItems": [
   {
     "areaPosition": [],
     "areaType": 0
   }
 ],
 "installMode": 0,
“detectBoundary”:[]
}
```

Params:

name	description	in	require	type	schema
Authorization	Api access token	header	true	string	
radarId	radarId	path	true	string	
roomConfigurationAreaDTO	RoomConfigurationAreaDTO	body	true	RoomConfigurationAreaDTO	RoomConfigurationAreaDTO

name	description	in	require	type	schema
areaItems	Area items, max support 5 items		false	array	AreaItem
areaPosition	XY[8] x1,y1,x2,y2,x3,y3, x4,y4 8 个 float-type codes that are arranged clockwise or counter-clockwise		true	array	float
areaType	Area type 0 :Door, 1:Sofa, 2:Bed		true	integer	
installMode	Install type 0: corner, 1: wall		true	integer(int32)	
detectBoundary	Detection Boundary: three float values between 0-7. 1. Wall-mount mode: reset to default val1=0 AND val2=0, OR val3=0. 2. Corner-mount mode: reset to default if:val1=0 OR val2=0. Only available for firmware version 2.8.1.44 and later..		false	array	float

Status:

code	description	schema
200	OK	restful api response object«Void»
201	Created	
401	Unauthorized	
403	Forbidden	

code	description	schema
404	Not Found	

Response Params:

name	description	type	schema
code	response code	string	
msg	response description message	string	

Response Example:

```
{
        "code": "200",
        "msg": "success"
}
```

3.44. To request for the room layout assigned to a specific radar (Assure)

url:/api/v1/room-configuration-area/{radarId}

method:GET

produces:application/x-www-form-urlencoded

consumes:*/*

Note:

Params:

name	description	in	require	type	schema
Authorization	Api access token	header	true	string	
radarId	radarId	path	true	string	

Status:

code	description	schema
200	OK	restful api response object«RoomConfigurationAreaDTO»
401	Unauthorized	
403	Forbidden	
404	Not Found	

Response Params:

name	description	type	schema
code	response code	string	
data	response data	RoomConfigurationAreaDTO	RoomConfigurationAreaDTO
areaItems	Area items, max support 5 items	array	AreaItem
areaPosition	XY[8] x1,y1,x2,y2,x3,y3,x4,y4 8 个 float-type codes that are arranged clockwise or counter-clockwise	array	float
areaType	Area type 0 :Door, 1:Sofa, 2:Bed	integer	
installMode	Install type 0: corner, 1: wall	integer(int32)	
detectBoundary	Detection Boundary: three float values between 0-7. 1. Wall-mount mode: reset to default val1=0 AND val2=0, OR val3=0. 2. Corner-mount mode: reset to default if:val1=0 OR val2=0. Only available for firmware version 2.8.1.44 and later.	array	float
msg	response description message	string	

Response Example:


```
{
        "code": "200",
        "data": {
                "areaItems": [
                        {
                                "areaPosition": [],
                                "areaType": 0
                        }
                ],
                "installMode": 0,
                “detectBoundary”:[]
        },
        "msg": "success"
}
```

3.45. To clear the room layout that is assigned to a specific radar (Assure)

url:/api/v1/room-configuration-area/clear/{radarId}

method:GET

produces:application/x-www-form-urlencoded

consumes:*/*

Note:

Params:

name	description	in	require	type	schema
Authorization	Api access token	header	true	string	
radarId	radarId	path	true	string	

Status:

code	description	schema
200	OK	restful api response object«Void»
401	Unauthorized	
403	Forbidden	
404	Not Found	

Response Params:

name	description	type	schema
code	response code	string	
msg	response description message	string	

Response Example:

```
{
        "code": "200",
        "msg": "success"
}
```

3.46. To set up the fall-detection mode of the radar (Assure)

url:/api/v1/fall-detection-mode

method:PUT

produces:application/x-www-form-urlencoded,application/json

consumes:*/*

Note: Command to set the fall detection mode

0: High sensitivity mode

1: Low false alert mode

Available to firmware of 2.8.x.x and later versions.

Example:

```
{
 "mode": 0,
 "radarId": ""
}
```

Params:

name	description	in	require	type	schema
Authorization	Api access token	header	true	string	
fallDetectionModeDTO	FallDetectionModeDTO	body	true	FallDetectionModeDTO	FallDetectionModeDTO
mode	Fall detection mode 0: High sensitivity mode 1: Low sensitivity mode		false	integer(int32)	
radarId	radar id		false	string	

Status:

code	description	schema
200	OK	restful api response object«Void»
201	Created	
401	Unauthorized	
403	Forbidden	
404	Not Found	

Response Params:

name	description	type	schema

name	description	type	schema
code	response code	string	
msg	response description message	string	

Response Example:


```
{
        "code": "200",
        "msg": "success"
}
```


3.47. To get the fall-detection mode of the radar (Assure)

url:/api/v1/fall-detection-mode/{radarId}

method:GET

produces:application/x-www-form-urlencoded

consumes:*/*

Note: Available to firmware of 2.8.x.x and later versions

Params:

name	description	in	require	type	schema
Authorization	Api access token	header	true	string	
radarId	radarId	path	true	string	

Status:

code	description	schema
200	OK	restful api response object«FallDetectionModeDTO»
401	Unauthorized	
403	Forbidden	
404	Not Found	

Response Params:

name	description	type	schema
code	response code	string	
data	response data	FallDetectionModeDTO	FallDetectionModeDTO
mode	Fall detection mode	integer(int32)	

name	description	type	schema
	0: High sensitivity mode 1: Low sensitivity mode		
radarId	radar id	string	
msg	response description message	string	

Response Example:


```
{
        "code": "200",
        "data": {
                "mode": 0,
                "radarId": ""
        },
        "msg": "success"
}
```


3.48. To get the push switch settings of the system (real-time coordinates of the target)

url:/api/v1/push-switches

method:GET

produces:application/x-www-form-urlencoded

consumes:*/*

Note: Get the current push switch configuration status of the system. The developer can use this interface to get the real-time coordinates of the moving target. Please note that the system support only one target.

Push switch list:

name	description	Push endpoint
PERSON_POSITION_DETECTION	Person position xyz data of Assure radar	/api/radar/person-position-detection

Params:

name	description	in	require	type	schema
Authorization	Api access token	header	true	string	

Status:

code	description	schema

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code	description	schema
200	OK	
401	Unauthorized	
403	Forbidden	
404	Not Found	

Response Params:

name	description	type	schema
code	response code	string	
data	response data	PushSwitches DTO	PushSwitchesDTO
pushSwitchesMap	Push switch mapping table, containing all configurable switches	Object	
msg	response description message	string	

Response Example:


```
{
        "code": "200",
        "data": {
                "pushSwitchesMap": {
                        "PERSON_POSITION_DETECTION":false
                }
        },
        "msg": "success"
}
```


3.49. To set the push switch settings of the system (real-time coordinates of the target)

url:/api/v1/push-switches

method:PUT

produces:application/x-www-form-urlencoded,application/json

consumes:*/*

Note: Configure the push switch status of the system. For specific configurable switches, please refer to the get push switch interface (Section 3.48).

Example:

```
{
    "pushSwitchesMap": {
        "PERSON_POSITION_DETECTION":true
    }
}
```

Params:

name	description	in	require	type	schema
Authorization	Api access token	header	true	string	
pushSwitchesDTO	PushSwitchesDTO	body	true	PushSwitchesDTO	PushSwitchesDTO
pushSwitchesMap	pushSwitchesMap Push switch mapping table: true: enable push false: disable push		false	Object	

Status:

code	description	schema
200	OK	restful api response object«Void»
201	Created	
401	Unauthorized	
403	Forbidden	
404	Not Found	

Response Params:

name	description	type	schema
code	response code	string	
msg	response description message	string	

Response Example:

```
{
        "code": "200",
        "msg": "success"
}
```

4. Support

Please contact support@ax-end.com for any help or more information.

AxEnd, Inc.
www.ax-end.com
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